

Week 5: Window Function

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Lab1: Film ranking

The screenshot shows a SQL query editor with a query window and a data output table. The query uses a window function to rank films by length within each category.

```
1  --Lab1
2  with ranking_table as(
3      select c.name, f.title, f.length,
4          rank() over(partition by c.name order by f.length desc, f.title ) as rank
5      from film f
6      join film_category fc on f.film_id=fc.film_id
7      join category c on c.category_id = fc.category_id
8  )
9  select * from ranking_table where rank<=3;
```

The data output table shows the results of the query, displaying the top 3 ranked films for each category. The table has 48 rows in total, but only the first 14 are visible in the screenshot.

	name	title	length	rank
	character varying (25)	character varying (255)	smallint	bigint
1	Action	Darn Forrester	185	1
2	Action	Worst Banger	185	2
3	Action	Casualties Encino	179	3
4	Animation	Gangs Pride	185	1
5	Animation	Pond Seattle	185	2
6	Animation	Sons Interview	184	3
7	Children	Fury Murder	178	1
8	Children	Wrong Behavior	178	2
9	Children	Empire Malkovich	177	3
10	Classics	Conspiracy Spirit	184	1
11	Classics	Dracula Crystal	176	2
12	Classics	Lights Deer	174	3
13	Comedy	Control Anthem	185	1
14	Comedy	Saturn Name	182	2

Total rows: 48 | Query complete 00:00:00.147 | CRLF | Ln 1, Col 7

Lab2: Growth Analysis

Query Query History

```

1  --Lab2]
2  with monthly_revenue as(
3      select
4          date_trunc('month', payment_date) as month,
5          sum(amount) as curr_revenue
6      from payment
7      group by(date_trunc('month', payment_date))
8  ),
9  revenue_with_lag as (
10     select month, curr_revenue,
11     lag (curr_revenue) over (order by month) as prev_revenue
12     from monthly_revenue
13 )
14
15     select month, curr_revenue, prev_revenue,
16     round(
17         (curr_revenue-prev_revenue)*100/prev_revenue,2
18     ) as MoM_Growth_Rate
19     from revenue_with_lag;
20

```

Data Output Messages Notifications

Showing rows: 1 to 4 Page No: 1 of 1

	month timestamp without time zone	curr_revenue numeric	prev_revenue numeric	mom_growth_rate numeric
1	2007-02-01 00:00:00	8351.84	[null]	[null]
2	2007-03-01 00:00:00	23886.56	8351.84	186.00
3	2007-04-01 00:00:00	28559.46	23886.56	19.56
4	2007-05-01 00:00:00	514.18	28559.46	-98.20

Total rows: 4 Query complete 00:00:00.096 CRLF Ln 1, Col 7

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Lab3: Customer Lifetime Value

Query Query History

```

1  select customer_id, amount, payment_date,
2  sum(amount) over(partition by customer_id order by payment_date desc) as Cumulative_Sum
3
4
5
6  from payment

```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 15

	customer_id smallint	amount numeric (5,2)	payment_date timestamp without time zone	cumulative_sum numeric
1	1	2.99	2007-04-30 01:10:44.996577	2.99
2	1	2.99	2007-04-29 02:27:15.996577	5.98
3	1	0.99	2007-04-28 17:48:33.996577	6.97
4	1	0.99	2007-04-28 16:02:05.996577	7.96
5	1	4.99	2007-04-28 14:46:49.996577	12.95
6	1	4.99	2007-04-28 07:33:11.996577	17.94
7	1	2.99	2007-04-27 09:59:48.996577	20.93
8	1	7.99	2007-04-11 08:42:12.996577	28.92
9	1	4.99	2007-04-09 15:06:27.996577	33.91
10	1	4.99	2007-04-09 11:52:33.996577	38.90
11	1	5.99	2007-04-08 06:02:22.996577	44.89
12	1	5.99	2007-04-08 01:45:31.996577	50.88
13	1	5.99	2007-03-22 18:32:12.996577	56.87
14	1	2.99	2007-03-22 18:10:03.996577	59.86
15	1	1.99	2007-03-21 23:56:23.996577	61.85
16	1	0.99	2007-03-21 22:02:23.996577	62.84

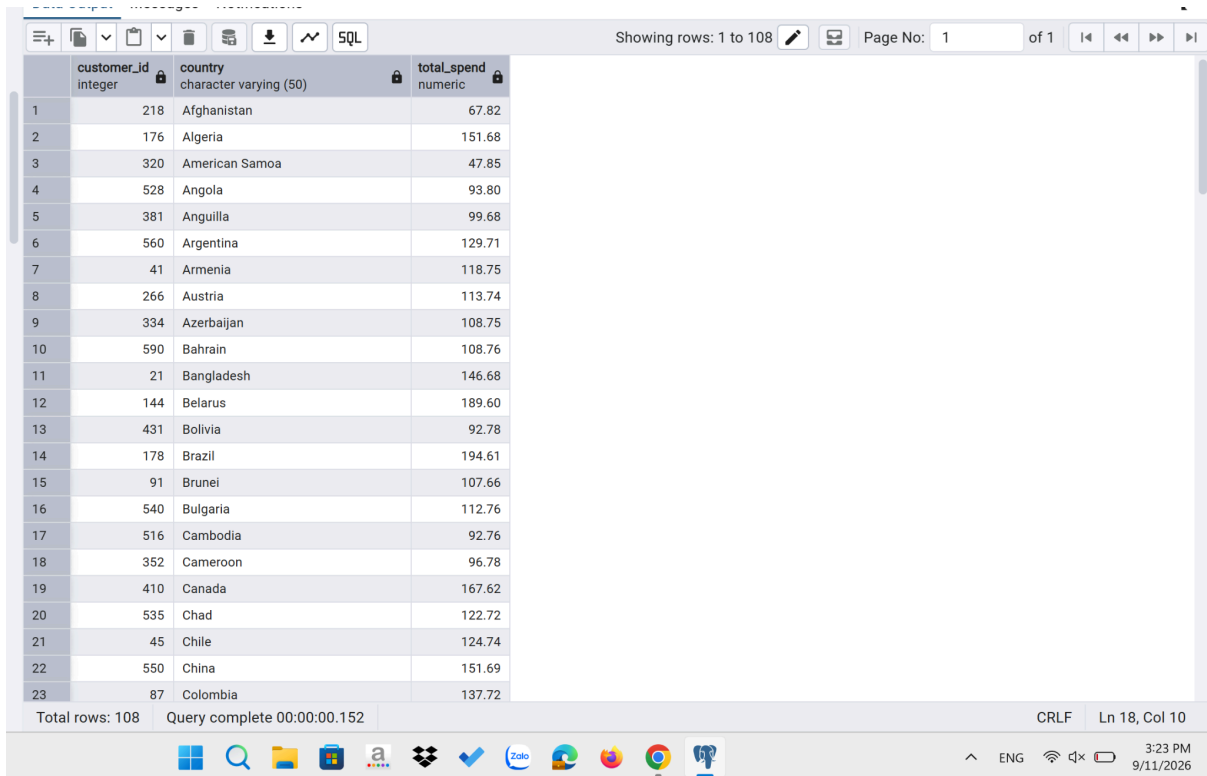
Total rows: 14596 Query complete 00:00:00.131 CRLF Ln 2, Col 11

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Lab4: The top N filter challenge

--Lab4

```
with customer_spend as (  
    select c.customer_id, cnt.country, sum(p.amount) as total_spend  
    from customer c  
    join address a on a.address_id=c.address_id  
    join city ct on ct.city_id=a.city_id  
    join country cnt on cnt.country_id=ct.country_id  
    join payment p on p.customer_id=c.customer_id  
    group by c.customer_id, cnt.country  
)  
  
rank_customers as (  
    select customer_id, country, total_spend,  
    rank() over(partition by country order by total_spend desc) as ranking  
    from customer_spend  
)  
  
select customer_id, country, total_spend  
from rank_customers  
where ranking=1;
```



Showing rows: 1 to 108 | Page No: 1 of 1

	customer_id integer	country character varying (50)	total_spend numeric
1	218	Afghanistan	67.82
2	176	Algeria	151.68
3	320	American Samoa	47.85
4	528	Angola	93.80
5	381	Anguilla	99.68
6	560	Argentina	129.71
7	41	Armenia	118.75
8	266	Austria	113.74
9	334	Azerbaijan	108.75
10	590	Bahrain	108.76
11	21	Bangladesh	146.68
12	144	Belarus	189.60
13	431	Bolivia	92.78
14	178	Brazil	194.61
15	91	Brunei	107.66
16	540	Bulgaria	112.76
17	516	Cambodia	92.76
18	352	Cameroon	96.78
19	410	Canada	167.62
20	535	Chad	122.72
21	45	Chile	124.74
22	550	China	151.69
23	87	Colombia	137.72

Total rows: 108 | Query complete 00:00:00.152 | CRLF | Ln 18, Col 10

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